02 · Lesson spec

Mini-project: Ayo's Ramp and Route · 40 minutes (can run over 2 sessions) · teams of 4 · fully unplugged · station check starts

Goal	**I can plan our build in 3 steps and guide Ayo with 3 cards, fixing a wrong card.**
Core parts	Code + Build
Key words	problem · plan · build · test · fix
Materials	Teacher-made ramp board or tower base per group; big blocks; large toy car; paper bread; plan cards; taped 3×3 grid; forward/stop cards; cut-out Ayo; build card and picture checklist (Printables); Station 1 kit (Guide 05)



1. Launch (5 min)

- **Ask the thinking steps:** "What is Ayo's problem?" (the bread car must get to the bottom of the ramp) "What small steps? Which way? What if it does not work?"
- Show the **build card** (6 steps) and the picture checklist: plan ✓ build and test ✓ route ✓ fix ✓.
- Give roles: **planner** (plan cards), **builder** (adds blocks), **card-layer** (Ayo's route), **checker** (points and checks).

2. Build and route (25 min)

- **Plan first (Core: Code):** the planner lays 3 plan cards: blocks → ramp → test, before anyone builds.
- **Build (Core):** add 2–3 big blocks to the teacher-made base to make the team's ramp or tower.
- **Test:** roll the toy car down the ramp (or stand the paper bread on the tower). If it falls off, **add a block and test again**.
- **Route (Core: Code):** the ramp sits at the end of a straight grid row. The card-layer lays forward, forward, stop; the checker points; push Ayo one square per card to the ramp. A partner swaps two cards; fix them and run again.
- Teams tick their picture checklist. The teacher **photos** each team with its build and route.
- **Stretch:** add one extra step or a second target.

Meanwhile: Station 1 (teacher, station check starts)

- Take **one small group at a time** (3–4 children, about 4 minutes). Each child does the 4 items (script in Guide 05): ① problem and small steps → ② another way → ③ change a build after a test → ④ order 3 cards and swap back two (MVL).
- Record E / D / S / X on the Station 1 record sheet. Aim for **8–12 children** this period. Continue in Week 12.

3. Show (10 min)

- Each team runs Ayo's route and tests its ramp for the class.
- **Create:** present: "Ayo's problem was… we solved it by…".
- If you can, show another class (walk there together with an adult).
- **Activity Book pp105–107:** the team build pages with a picture checklist. Children colour each step as their team does it and draw their team's build. Not marked.

Common mistakes

- Building before planning → "Plan cards first: blocks, ramp, test."
- Giving up when the car falls off → "Great test! Add a block and test again."
- Moving Ayo two squares for one card → "One card, one square."

What if it doesn't work?

- Not enough time → build in session 1, route and show in session 2.
- No toy car → a large plastic bottle rolls well.

Support and stretch

- **Support:** the base is nearly finished; the team adds one block and tests. The teacher lays the first route card.
- **Stretch** (optional, never assessed): one extra step or a second target.

How we check

Observation, picture build checklist and photo (not marked). Station 1 record sheet for the station check.

> **Safety**
> Big blocks only; adults cut; no climbing on builds; towers no taller than a knee; car wheels checked; walk with an adult to visit another class.